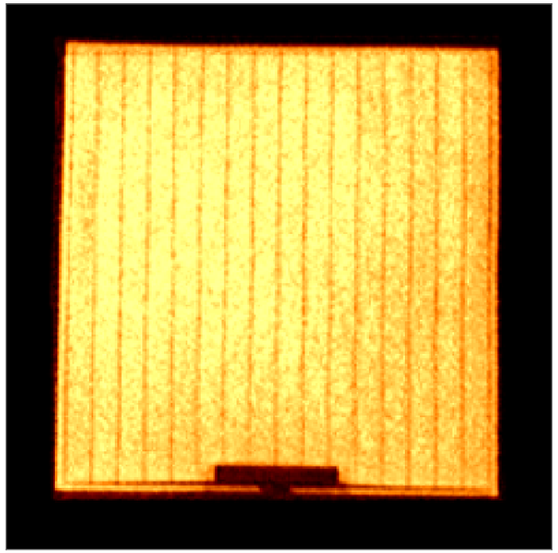
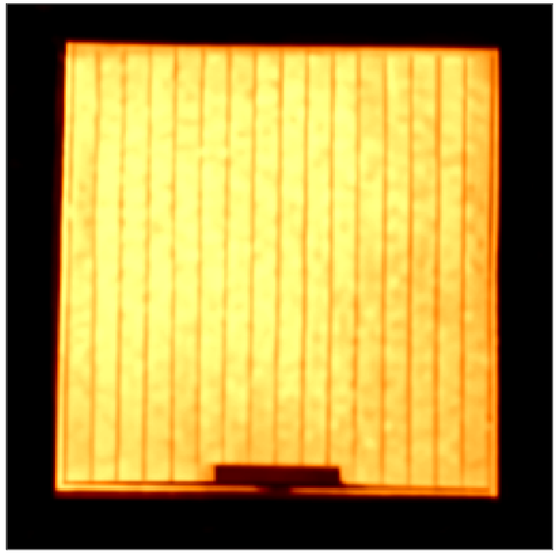
Bad-UTV evaluation | Reference_0021

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

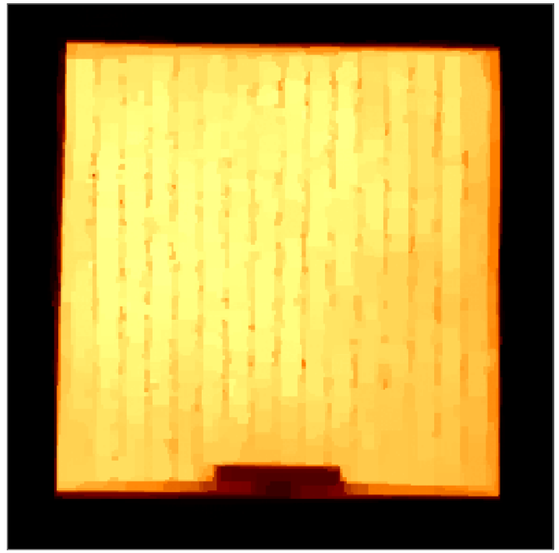
GT
Ground truth



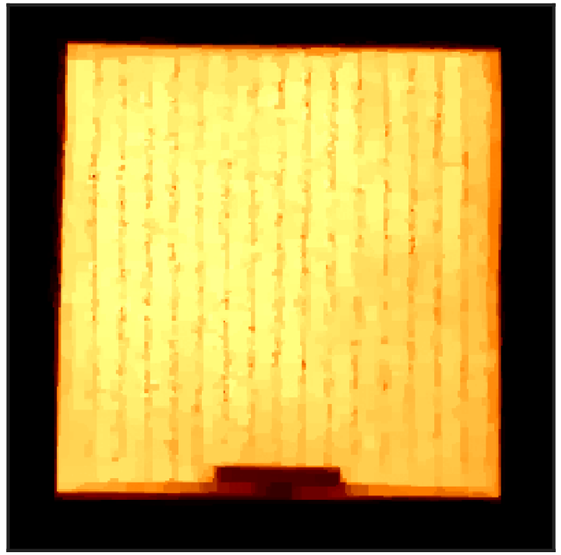
PnP-ADMM-tuned-5
PSNR 31.07 dB | SSIM 0.847
HaarPSI 0.822



C60 (rho=1)
PSNR 27.83 dB | SSIM 0.695
HaarPSI 0.574



C60 optimum (rho=0.63167)
PSNR 28.03 dB | SSIM 0.732
HaarPSI 0.632



Per-image PSNR gain from scaling: +0.200 dB | Compared with PnP-ADMM-tuned-5: -3.041 dB

Ternary rho range [0.0001, 10]; final bracket [0.62829, 0.63505], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

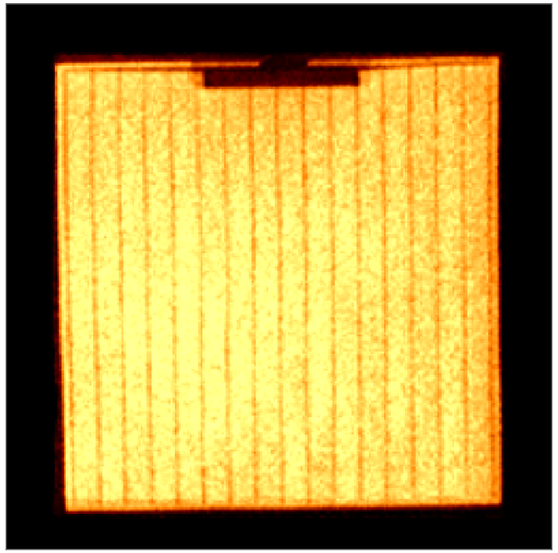


Display [0,1]; metrics on raw images

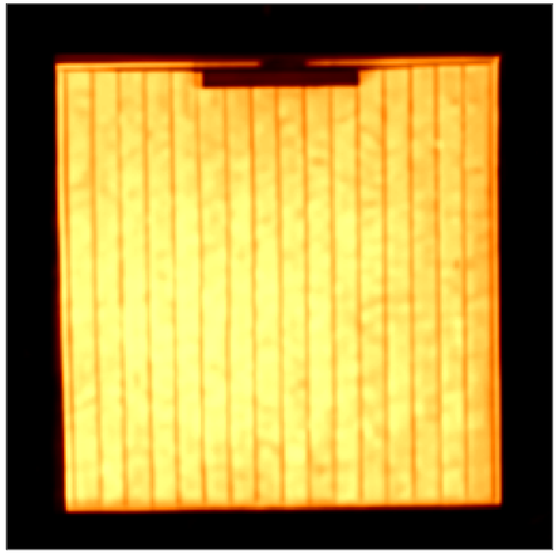
Bad-UTV evaluation | Reference_0066

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

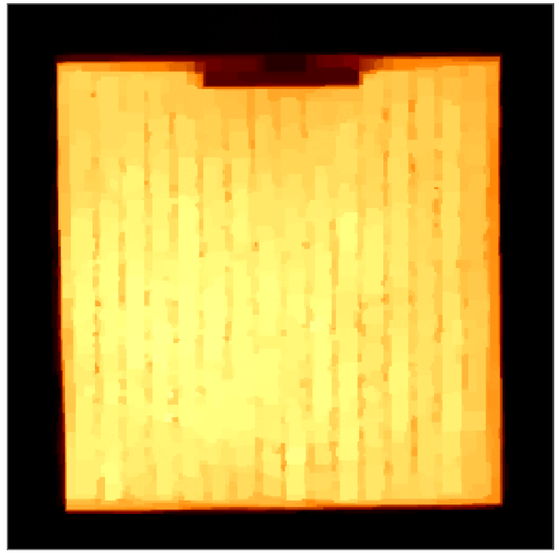
GT
Ground truth



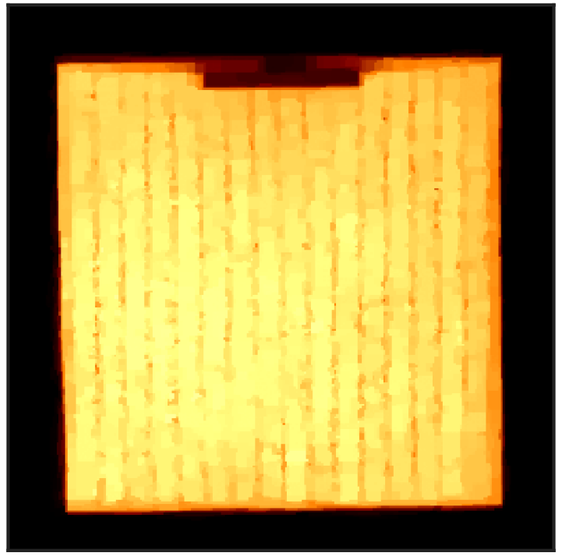
PnP-ADMM-tuned-5
PSNR 31.05 dB | SSIM 0.846
HaarPSI 0.814



C60 (rho=1)
PSNR 27.67 dB | SSIM 0.688
HaarPSI 0.568



C60 optimum (rho=0.57714)
PSNR 27.94 dB | SSIM 0.735
HaarPSI 0.635



Per-image PSNR gain from scaling: +0.272 dB | Compared with PnP-ADMM-tuned-5: -3.113 dB

Ternary rho range [0.0001, 10]; final bracket [0.57375, 0.58052], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

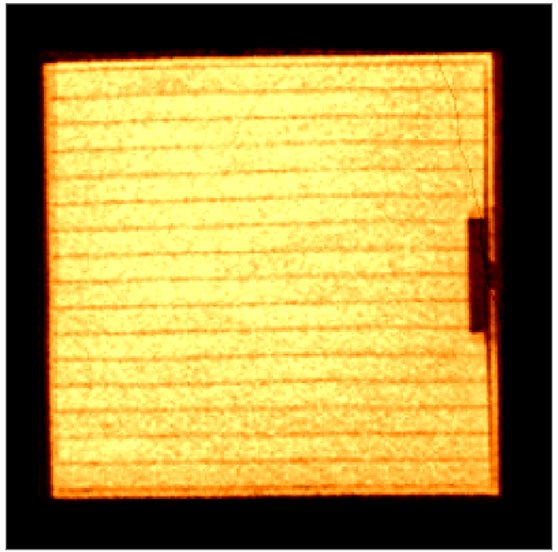


Display [0,1]; metrics on raw images

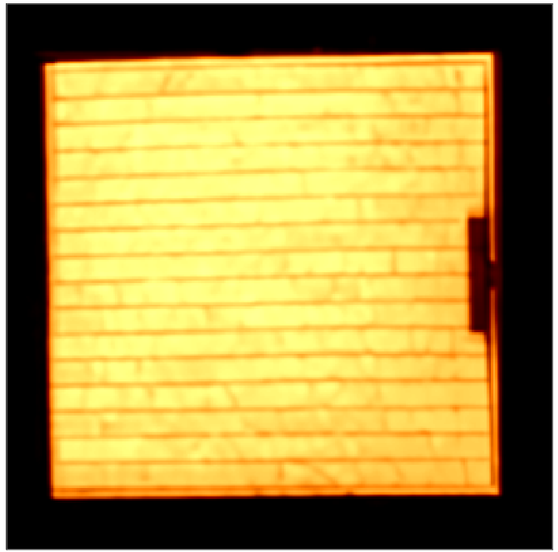
Bad-UTV evaluation | Reference_0188

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

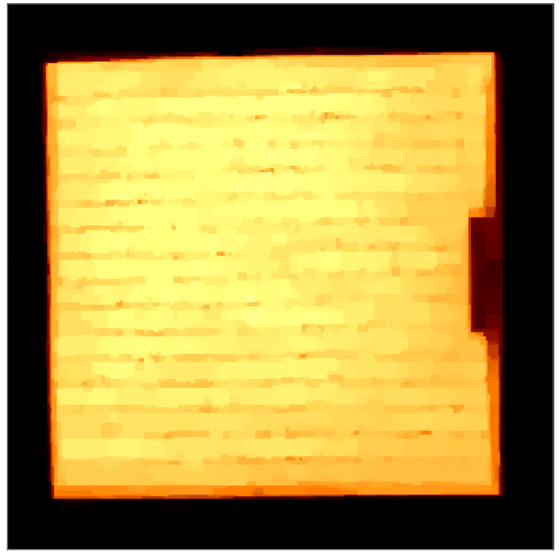
GT
Ground truth



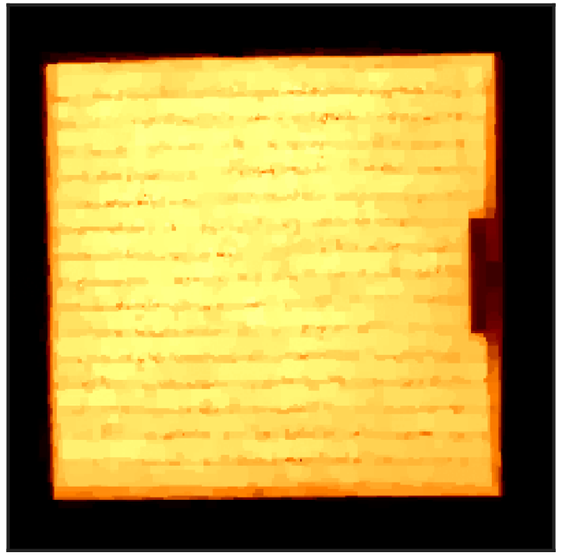
PnP-ADMM-tuned-5
PSNR 30.84 dB | SSIM 0.838
HaarPSI 0.818



C60 (rho=1)
PSNR 27.74 dB | SSIM 0.700
HaarPSI 0.596



C60 optimum (rho=0.63916)
PSNR 27.93 dB | SSIM 0.736
HaarPSI 0.655



Per-image PSNR gain from scaling: +0.192 dB | Compared with PnP-ADMM-tuned-5: -2.906 dB

Ternary rho range [0.0001, 10]; final bracket [0.63577, 0.64254], width 0.00677; 18 iterations, 51 evaluations

UTV C60 loses fine details or contrast

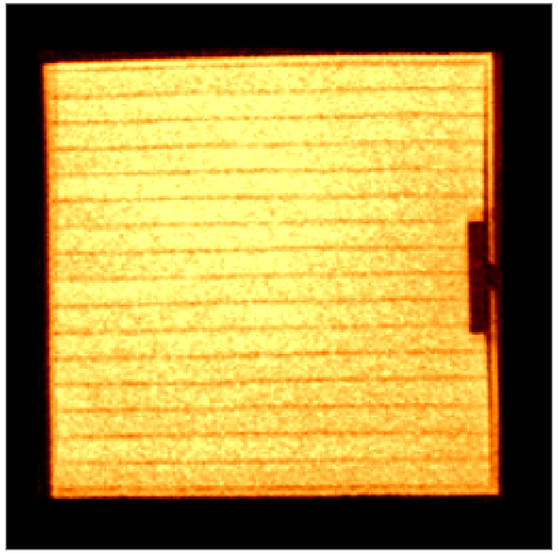


Display [0,1]; metrics on raw images

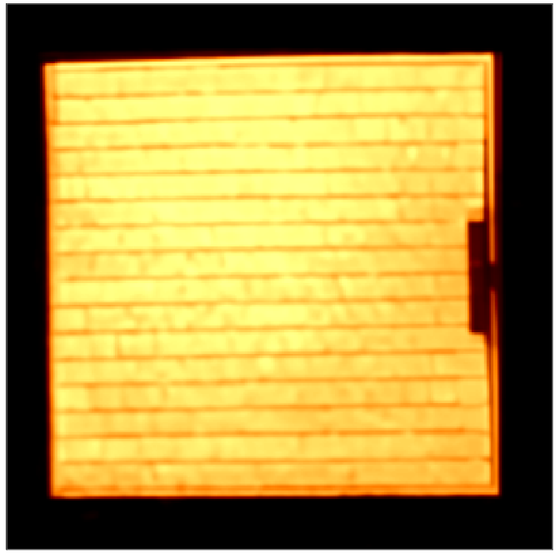
Bad-UTV evaluation | Reference_0199

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

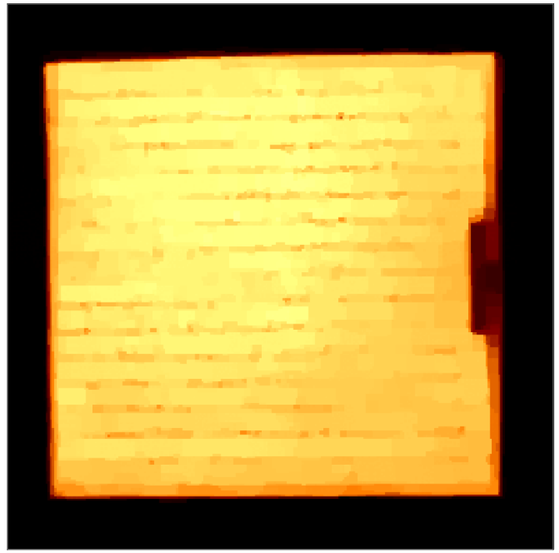
GT
Ground truth



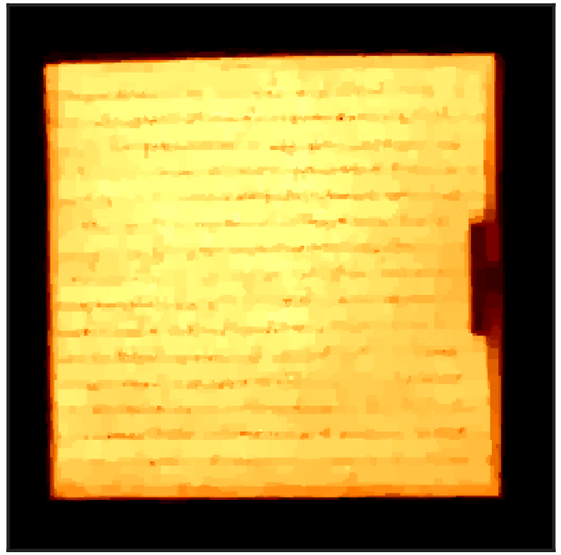
PnP-ADMM-tuned-5
PSNR 30.96 dB | SSIM 0.839
HaarPSI 0.819



C60 (rho=1)
PSNR 27.77 dB | SSIM 0.688
HaarPSI 0.579



C60 optimum (rho=0.64254)
PSNR 27.97 dB | SSIM 0.722
HaarPSI 0.644



Per-image PSNR gain from scaling: +0.192 dB | Compared with PnP-ADMM-tuned-5: -2.990 dB

Ternary rho range [0.0001, 10]; final bracket [0.63746, 0.64423], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

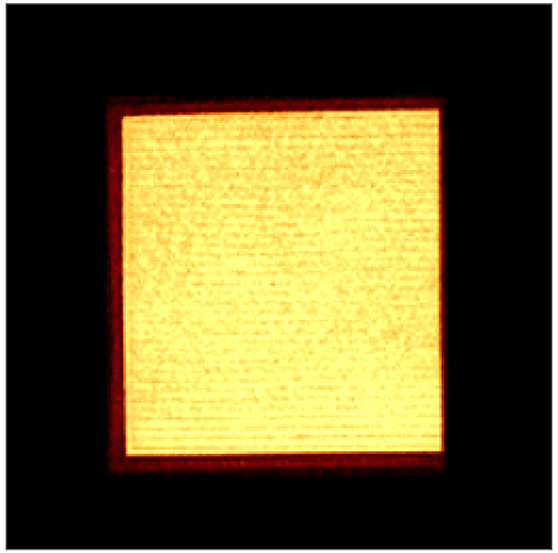


Display [0,1]; metrics on raw images

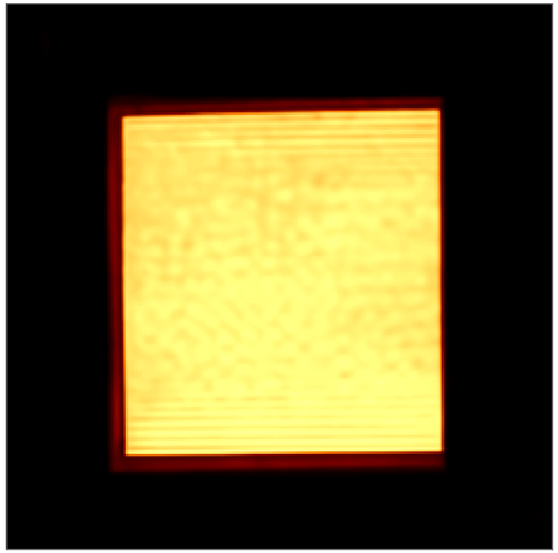
Bad-UTV evaluation | Reference_0260

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

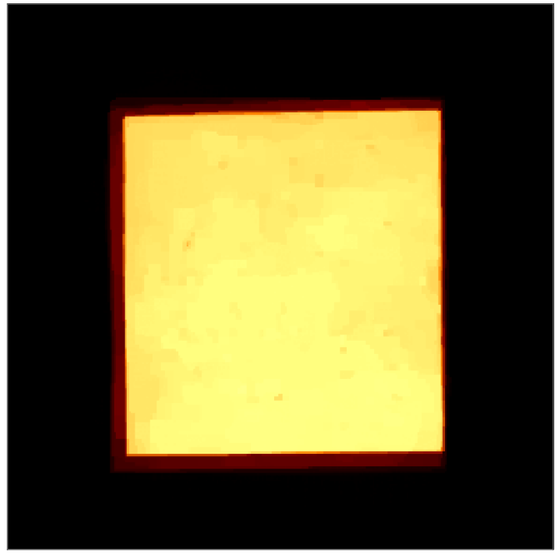
GT
Ground truth



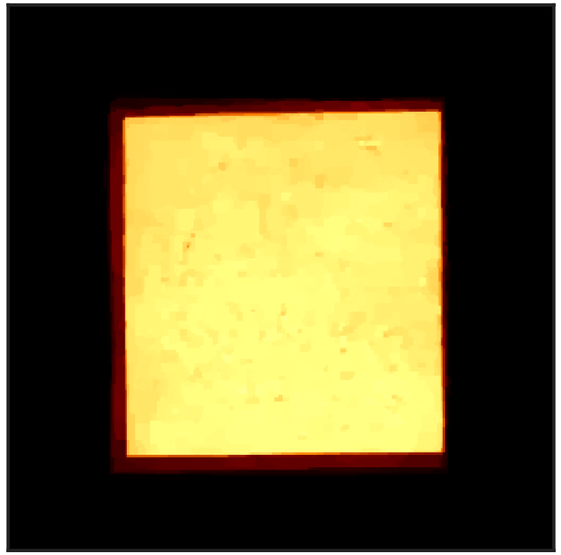
PnP-ADMM-tuned-5
PSNR 33.76 dB | SSIM 0.860
HaarPSI 0.825



C60 (rho=1)
PSNR 32.38 dB | SSIM 0.791
HaarPSI 0.650



C60 optimum (rho=0.69970)
PSNR 32.44 dB | SSIM 0.813
HaarPSI 0.688



Per-image PSNR gain from scaling: +0.066 dB | Compared with PnP-ADMM-tuned-5: -1.319 dB

Ternary rho range [0.0001, 10]; final bracket [0.69631, 0.70308], width 0.00677; 18 iterations, 51 evaluations

Visual observation: all compared methods failed to recover the lines.

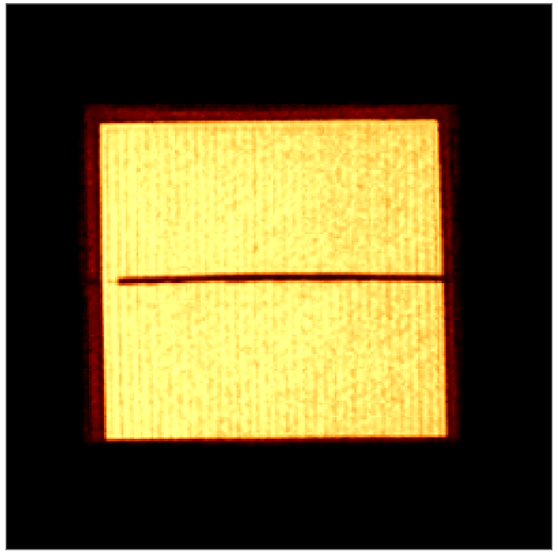


Display [0,1]; metrics on raw images

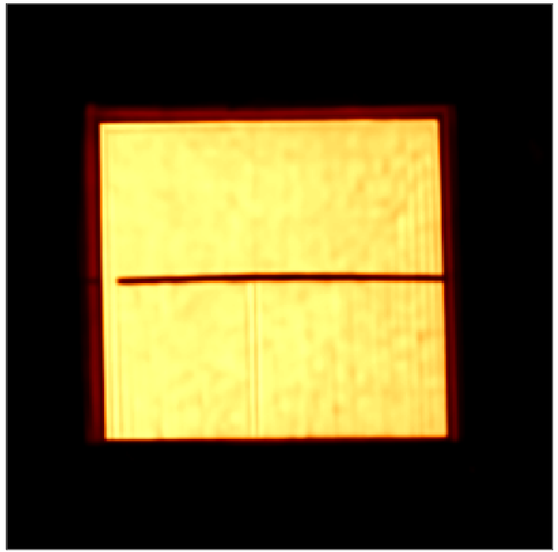
Bad-UTV evaluation | Reference_0357

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

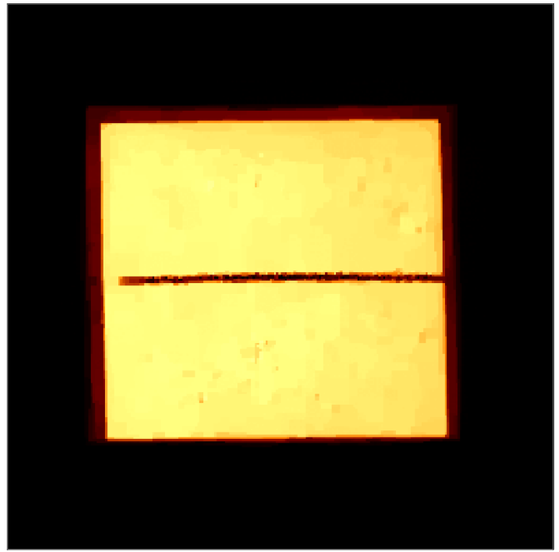
GT
Ground truth



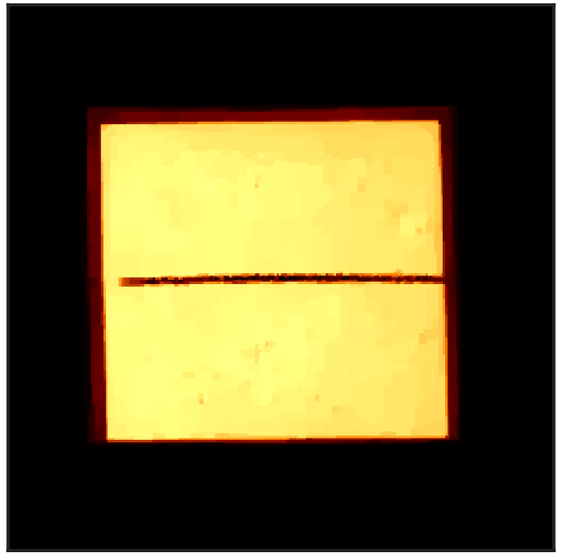
PnP-ADMM-tuned-5
PSNR 33.46 dB | SSIM 0.864
HaarPSI 0.806



C60 (rho=1)
PSNR 31.50 dB | SSIM 0.802
HaarPSI 0.685



C60 optimum (rho=1.16787)
PSNR 31.52 dB | SSIM 0.789
HaarPSI 0.663



Per-image PSNR gain from scaling: +0.019 dB | Compared with PnP-ADMM-tuned-5: -1.944 dB

Ternary rho range [0.0001, 10]; final bracket [1.16448, 1.17125], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

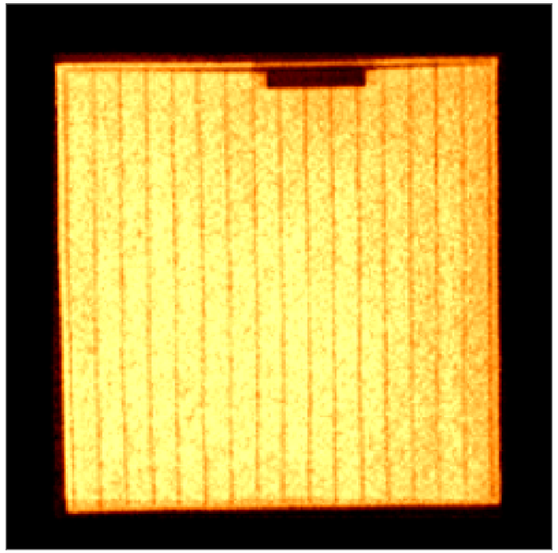


Display [0,1]; metrics on raw images

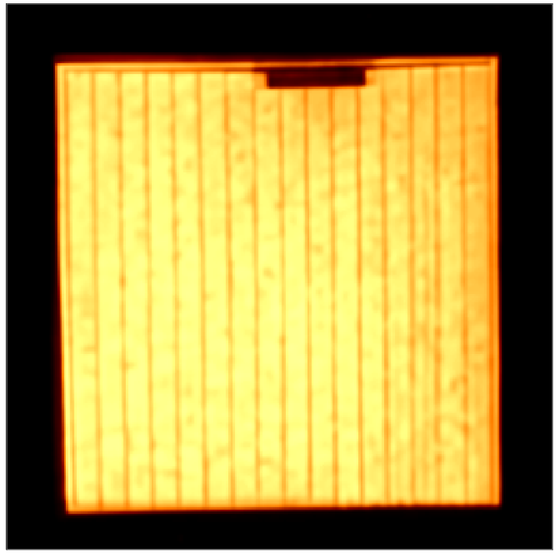
Bad-UTV evaluation | Reference_0727

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

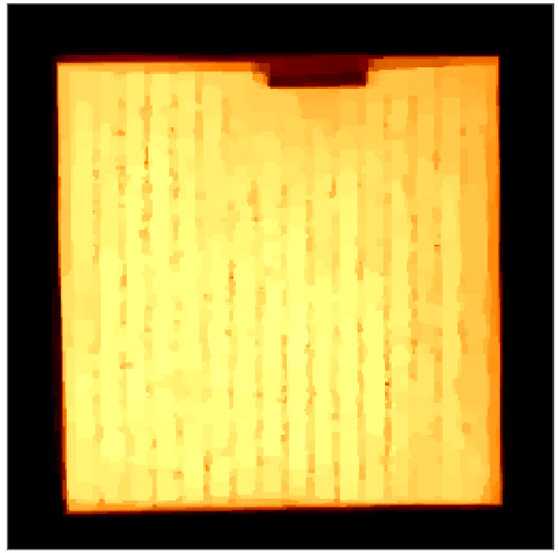
GT
Ground truth



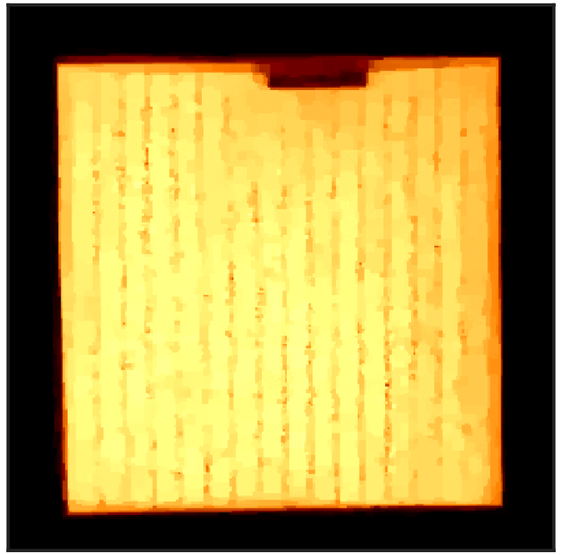
PnP-ADMM-tuned-5
PSNR 31.04 dB | SSIM 0.845
HaarPSI 0.828



C60 (rho=1)
PSNR 27.83 dB | SSIM 0.698
HaarPSI 0.600



C60 optimum (rho=0.68871)
PSNR 27.94 dB | SSIM 0.727
HaarPSI 0.640



Per-image PSNR gain from scaling: +0.111 dB | Compared with PnP-ADMM-tuned-5: -3.095 dB

Ternary rho range [0.0001, 10]; final bracket [0.68533, 0.69210], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

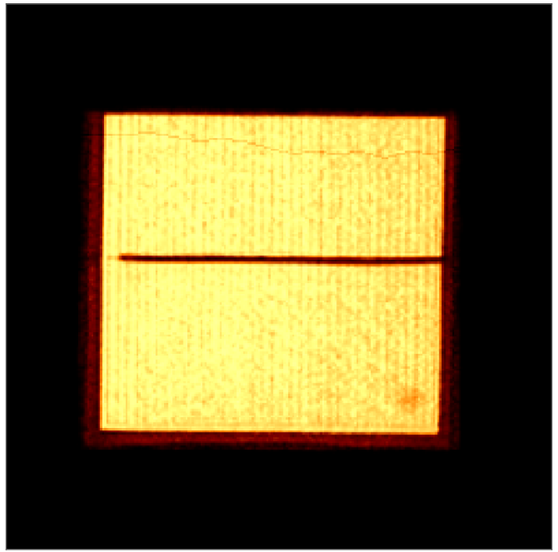


Display [0,1]; metrics on raw images

Bad-UTV evaluation | Reference_0793

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

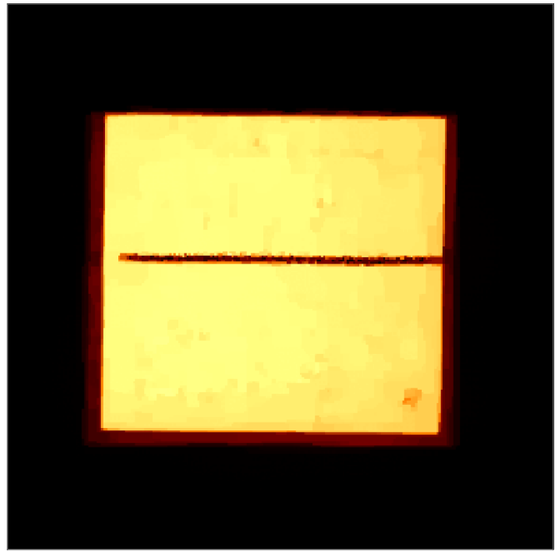
GT
Ground truth



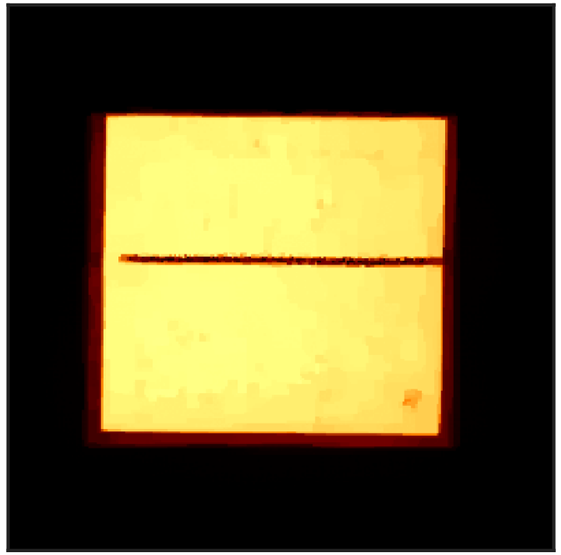
PnP-ADMM-tuned-5
PSNR 33.17 dB | SSIM 0.854
HaarPSI 0.778



C60 (rho=1)
PSNR 31.46 dB | SSIM 0.782
HaarPSI 0.683



C60 optimum (rho=1.12155)
PSNR 31.47 dB | SSIM 0.773
HaarPSI 0.673



Per-image PSNR gain from scaling: +0.009 dB | Compared with PnP-ADMM-tuned-5: -1.695 dB

Ternary rho range [0.0001, 10]; final bracket [1.11817, 1.12494], width 0.00677; 18 iterations, 50 evaluations

Visual observation: all compared methods failed.

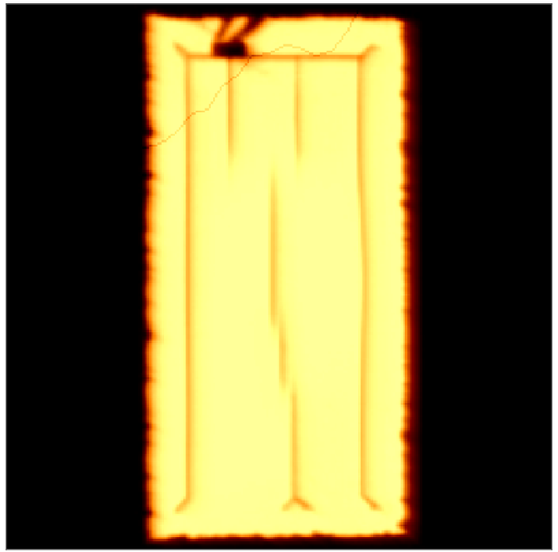


Display [0,1]; metrics on raw images

Bad-UTV evaluation | CIGS_0254

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

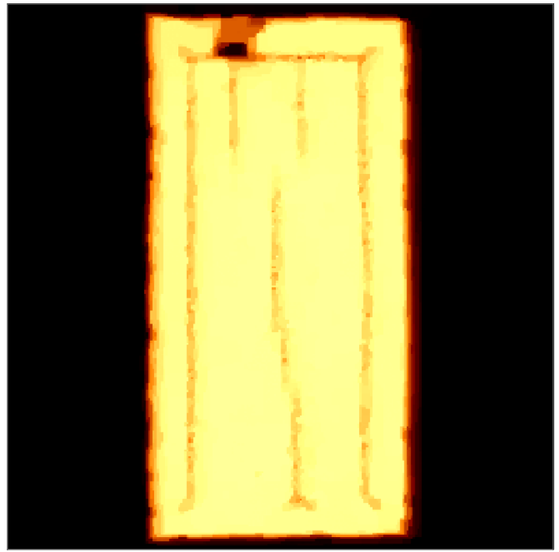
GT
Ground truth



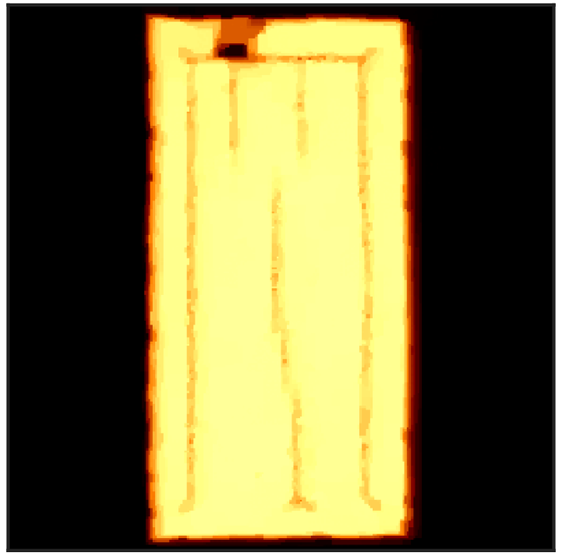
PnP-ADMM-tuned-5
PSNR 39.50 dB | SSIM 0.983
HaarPSI 0.947



C60 (rho=1)
PSNR 33.34 dB | SSIM 0.938
HaarPSI 0.825



C60 optimum (rho=1.09990)
PSNR 33.35 dB | SSIM 0.936
HaarPSI 0.818



Per-image PSNR gain from scaling: +0.010 dB | Compared with PnP-ADMM-tuned-5: -6.142 dB

Ternary rho range [0.0001, 10]; final bracket [1.09651, 1.10328], width 0.00677; 18 iterations, 52 evaluations

UTV C60 loses fine details or contrast



Display [0,1]; metrics on raw images

Bad-UTV evaluation | CIGS_0710

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

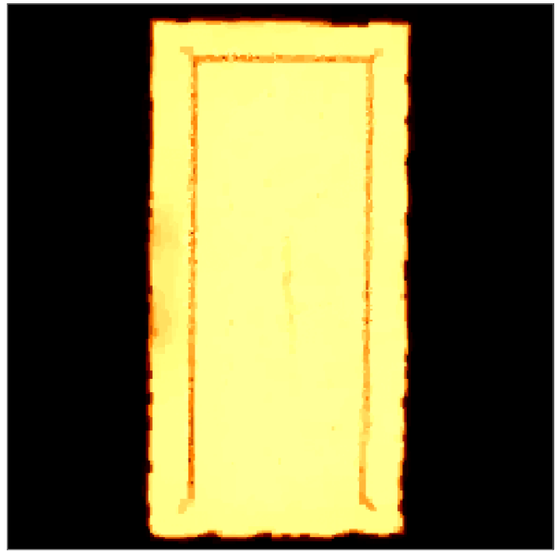
GT
Ground truth



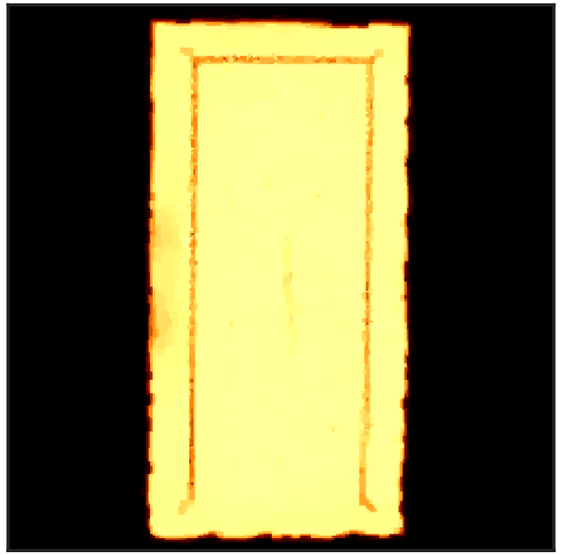
PnP-ADMM-tuned-5
PSNR 37.25 dB | SSIM 0.985
HaarPSI 0.937



C60 (rho=1)
PSNR 32.60 dB | SSIM 0.957
HaarPSI 0.875



C60 optimum (rho=0.97724)
PSNR 32.60 dB | SSIM 0.957
HaarPSI 0.876



Per-image PSNR gain from scaling: +0.002 dB | Compared with PnP-ADMM-tuned-5: -4.654 dB

Ternary rho range [0.0001, 10]; final bracket [0.97555, 0.98231], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast



Display [0,1]; metrics on raw images

Bad-UTV evaluation | CIGS_0727

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

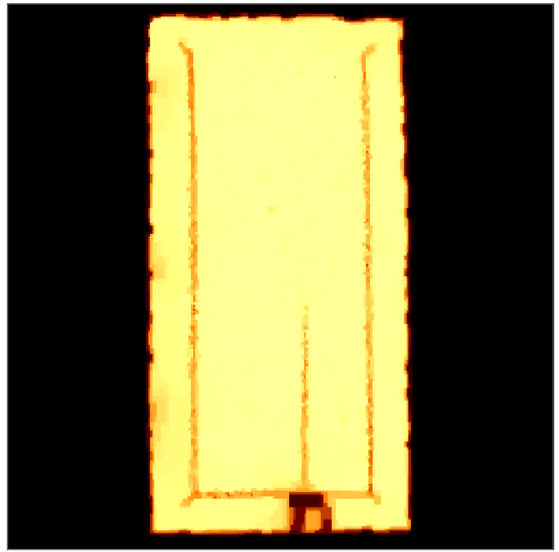
GT
Ground truth



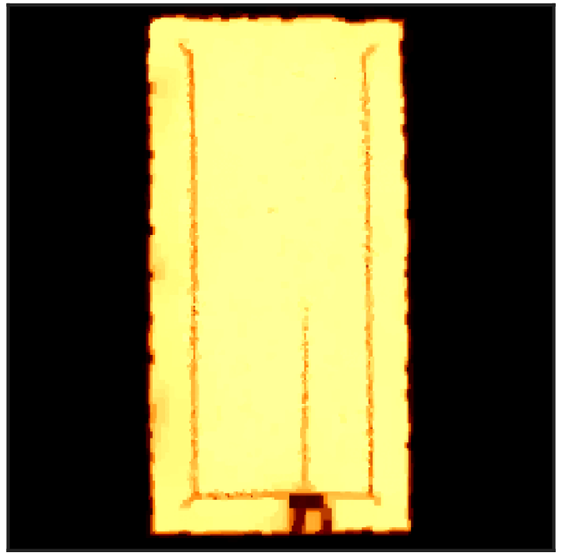
PnP-ADMM-tuned-5
PSNR 36.98 dB | SSIM 0.978
HaarPSI 0.927



C60 (rho=1)
PSNR 31.51 dB | SSIM 0.945
HaarPSI 0.856



C60 optimum (rho=0.87681)
PSNR 31.55 dB | SSIM 0.943
HaarPSI 0.863



Per-image PSNR gain from scaling: +0.040 dB | Compared with PnP-ADMM-tuned-5: -5.431 dB

Ternary rho range [0.0001, 10]; final bracket [0.87342, 0.88019], width 0.00677; 18 iterations, 51 evaluations

UTV C60 loses fine details or contrast



Display [0,1]; metrics on raw images

Bad-UTV evaluation | CIGS_0805

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

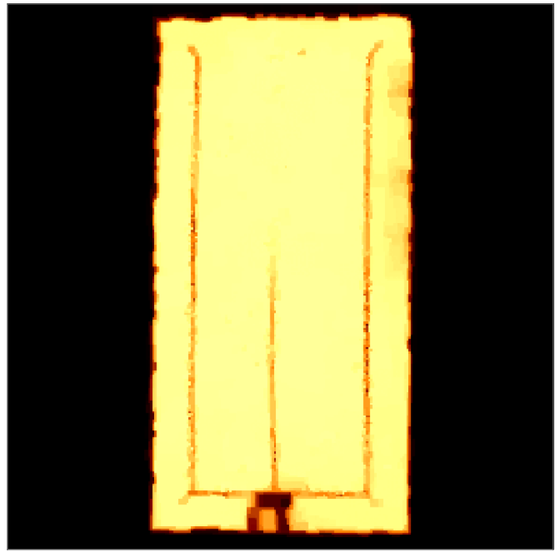
GT
Ground truth



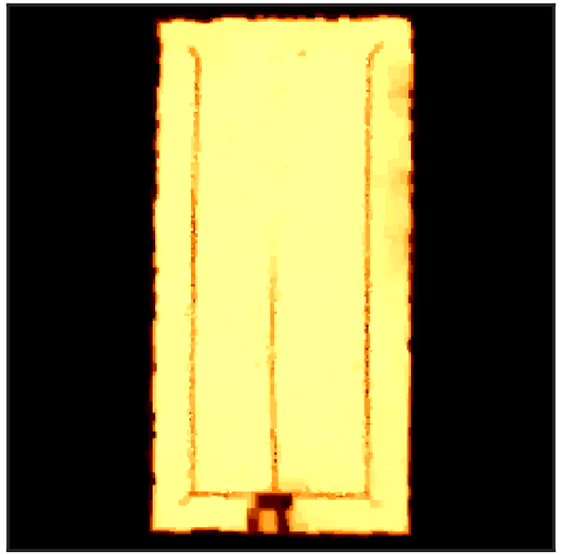
PnP-ADMM-tuned-5
PSNR 36.81 dB | SSIM 0.982
HaarPSI 0.926



C60 (rho=1)
PSNR 31.65 dB | SSIM 0.951
HaarPSI 0.868



C60 optimum (rho=0.93371)
PSNR 31.66 dB | SSIM 0.950
HaarPSI 0.871



Per-image PSNR gain from scaling: +0.013 dB | Compared with PnP-ADMM-tuned-5: -5.149 dB

Ternary rho range [0.0001, 10]; final bracket [0.93033, 0.93709], width 0.00677; 18 iterations, 51 evaluations

UTV C60 loses fine details or contrast

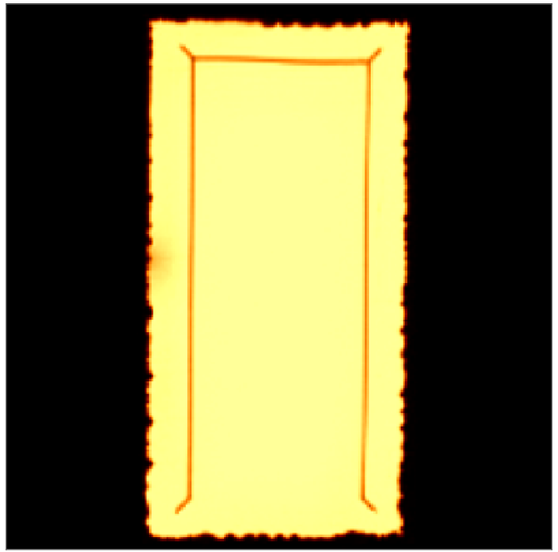


Display [0,1]; metrics on raw images

Bad-UTV evaluation | CIGS_0821

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

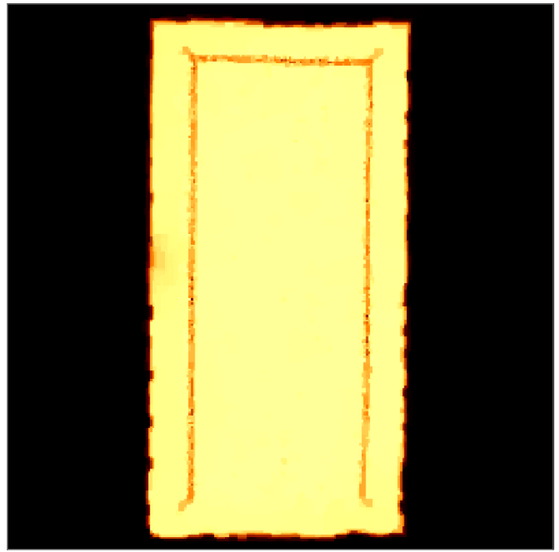
GT
Ground truth



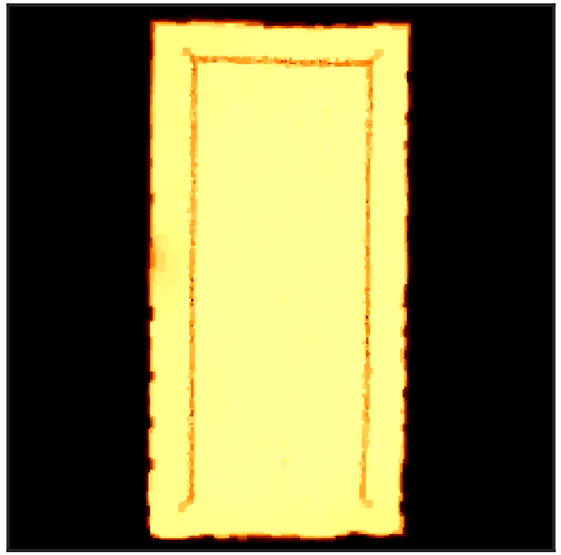
PnP-ADMM-tuned-5
PSNR 37.80 dB | SSIM 0.987
HaarPSI 0.938



C60 (rho=1)
PSNR 32.53 dB | SSIM 0.960
HaarPSI 0.892



C60 optimum (rho=1.11225)
PSNR 32.55 dB | SSIM 0.962
HaarPSI 0.887



Per-image PSNR gain from scaling: +0.017 dB | Compared with PnP-ADMM-tuned-5: -5.257 dB

Ternary rho range [0.0001, 10]; final bracket [1.10717, 1.11394], width 0.00677; 18 iterations, 52 evaluations

UTV C60 loses fine details or contrast

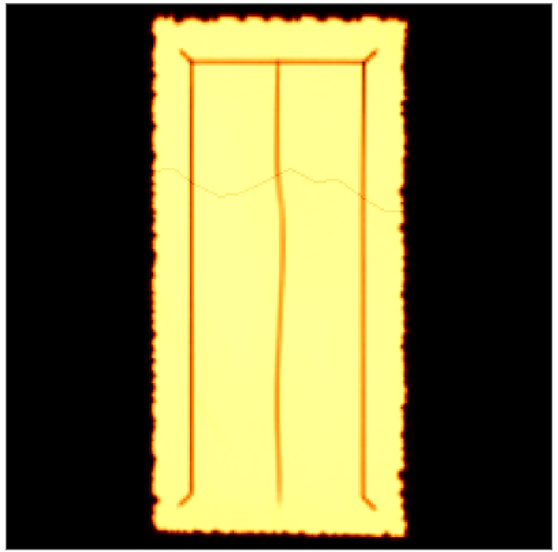


Display [0,1]; metrics on raw images

Bad-UTV evaluation | CIGS_0873

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

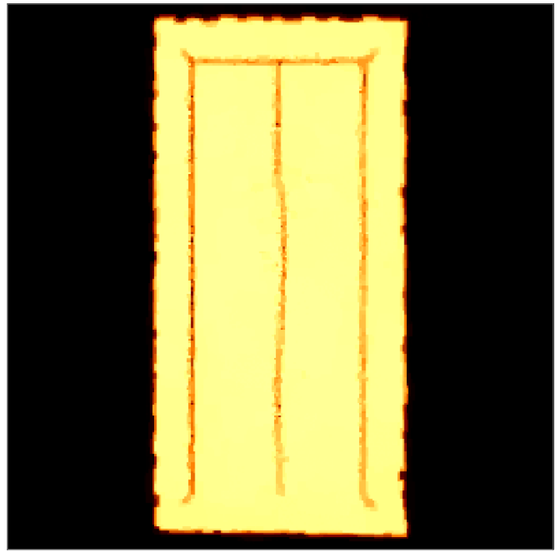
GT
Ground truth



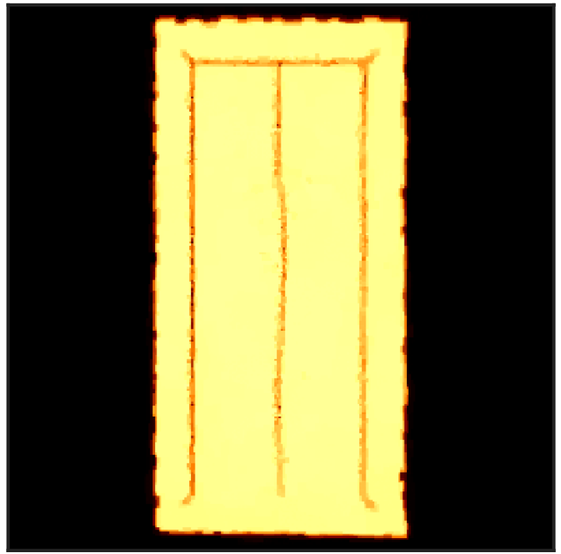
PnP-ADMM-tuned-5
PSNR 37.82 dB | SSIM 0.979
HaarPSI 0.946



C60 (rho=1)
PSNR 31.32 dB | SSIM 0.948
HaarPSI 0.857



C60 optimum (rho=0.93729)
PSNR 31.33 dB | SSIM 0.948
HaarPSI 0.859



Per-image PSNR gain from scaling: +0.010 dB | Compared with PnP-ADMM-tuned-5: -6.490 dB

Ternary rho range [0.0001, 10]; final bracket [0.93391, 0.94067], width 0.00677; 18 iterations, 51 evaluations

UTV C60 loses fine details or contrast

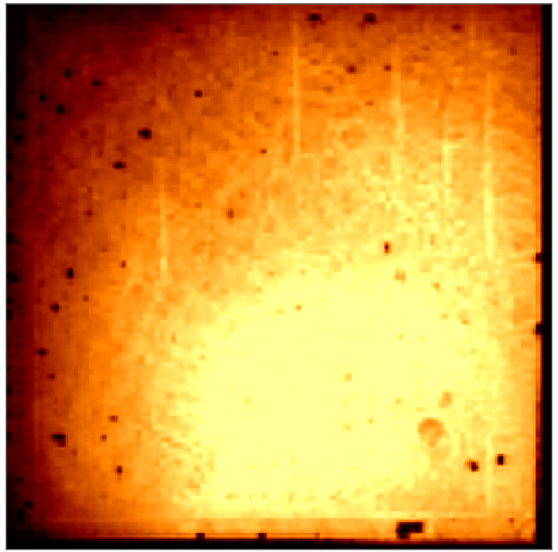


Display [0,1]; metrics on raw images

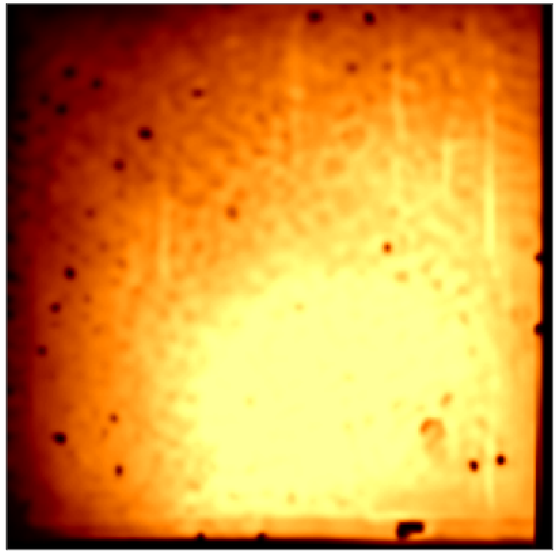
Bad-UTV evaluation | Perovskite_0021

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

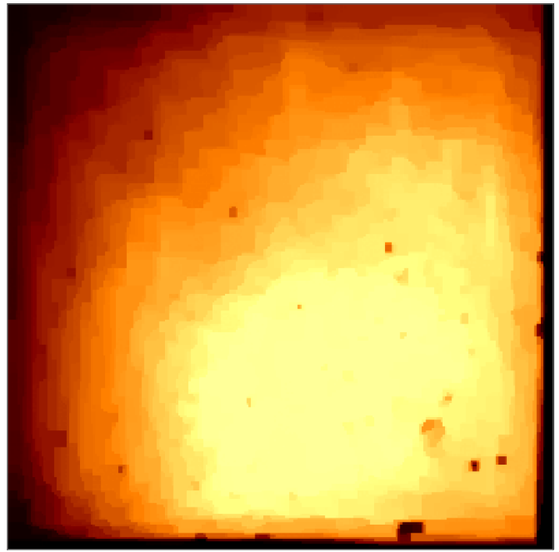
GT
Ground truth



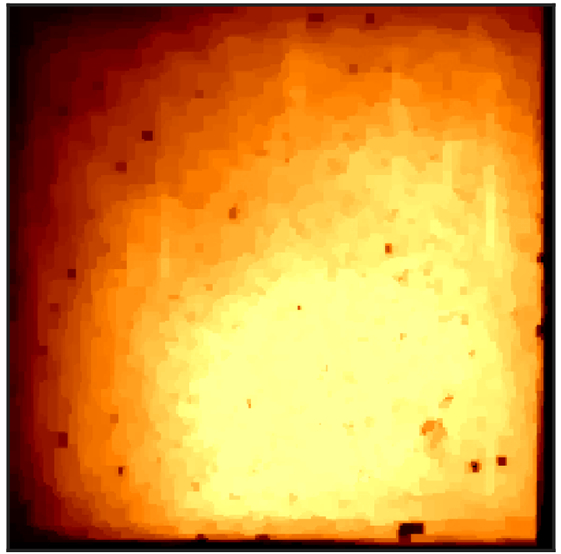
PnP-ADMM-tuned-5
PSNR 33.50 dB | SSIM 0.866
HaarPSI 0.801



C60 (rho=1)
PSNR 30.95 dB | SSIM 0.796
HaarPSI 0.623



C60 optimum (rho=0.49284)
PSNR 31.63 dB | SSIM 0.808
HaarPSI 0.706



Per-image PSNR gain from scaling: +0.679 dB | Compared with PnP-ADMM-tuned-5: -1.870 dB

Ternary rho range [0.0001, 10]; final bracket [0.48945, 0.49622], width 0.00677; 18 iterations, 51 evaluations

UTV C60 loses fine details or contrast

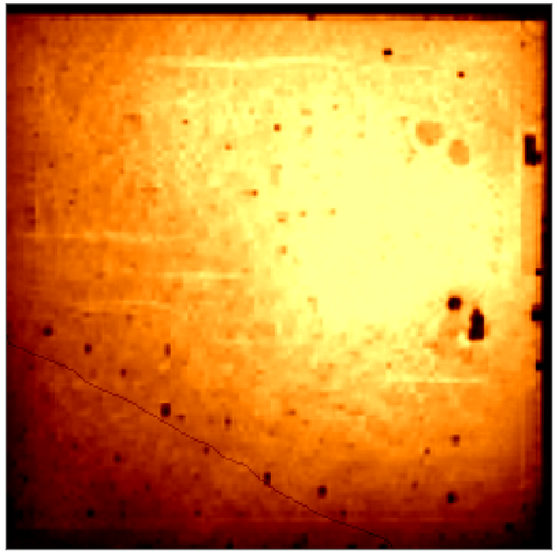


Display [0,1]; metrics on raw images

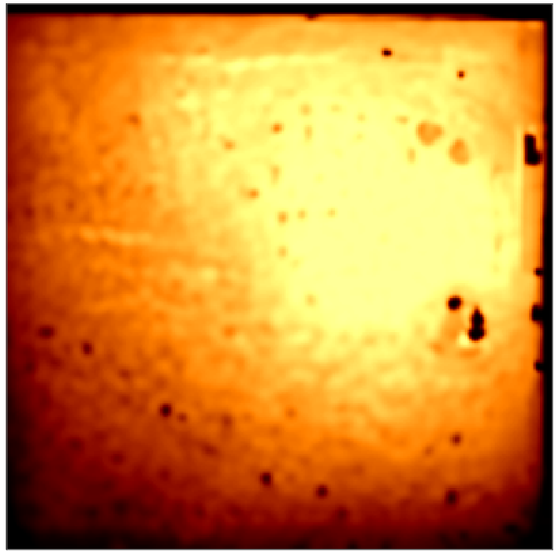
Bad-UTV evaluation | Perovskite_0031

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

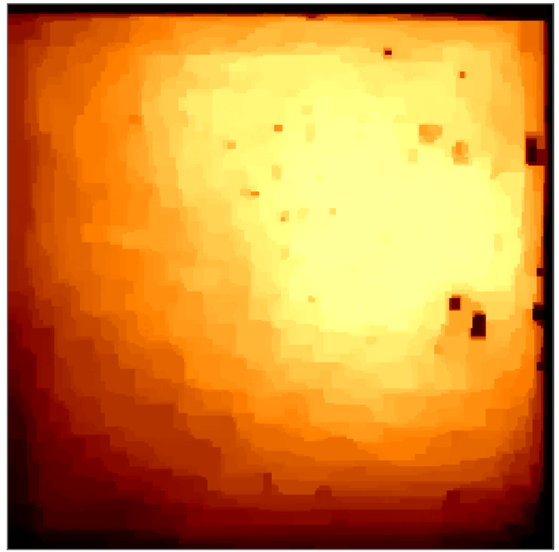
GT
Ground truth



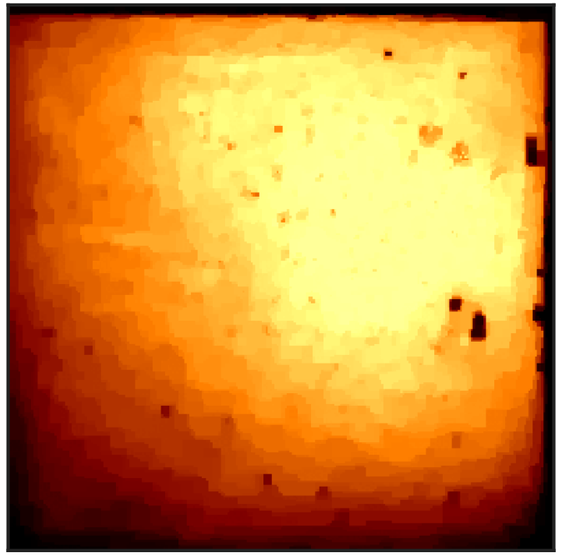
PnP-ADMM-tuned-5
PSNR 33.00 dB | SSIM 0.849
HaarPSI 0.789



C60 (rho=1)
PSNR 30.46 dB | SSIM 0.780
HaarPSI 0.615



C60 optimum (rho=0.43241)
PSNR 31.16 dB | SSIM 0.793
HaarPSI 0.700



Per-image PSNR gain from scaling: +0.698 dB | Compared with PnP-ADMM-tuned-5: -1.847 dB

Ternary rho range [0.0001, 10]; final bracket [0.43072, 0.43748], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

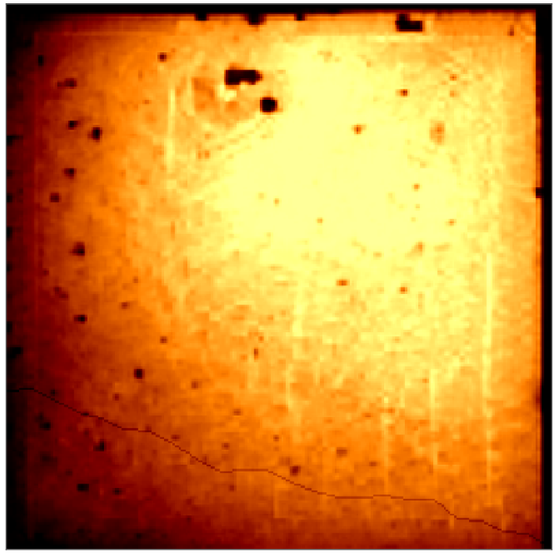


Display [0,1]; metrics on raw images

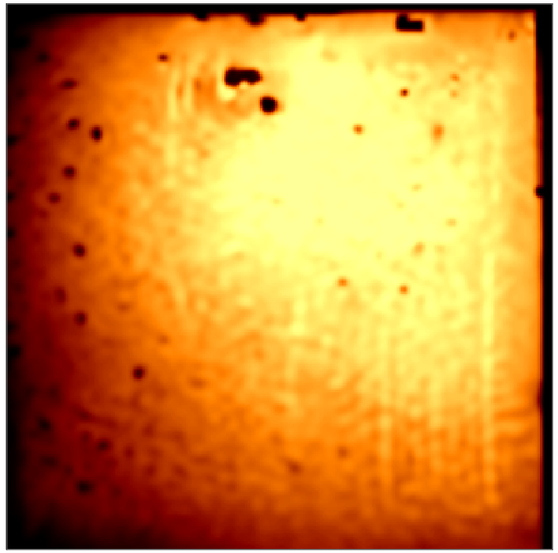
Bad-UTV evaluation | Perovskite_0079

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

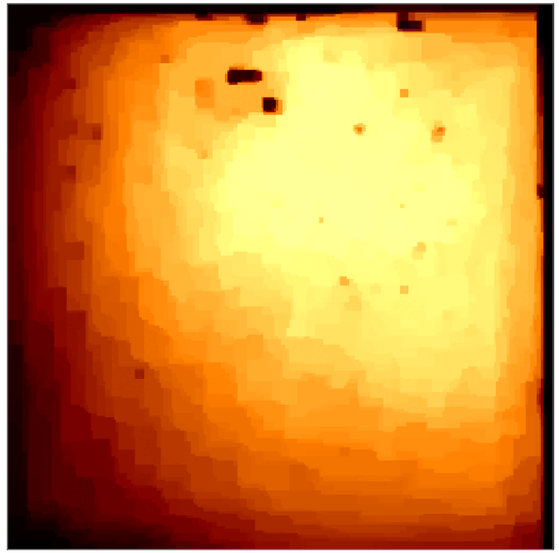
GT
Ground truth



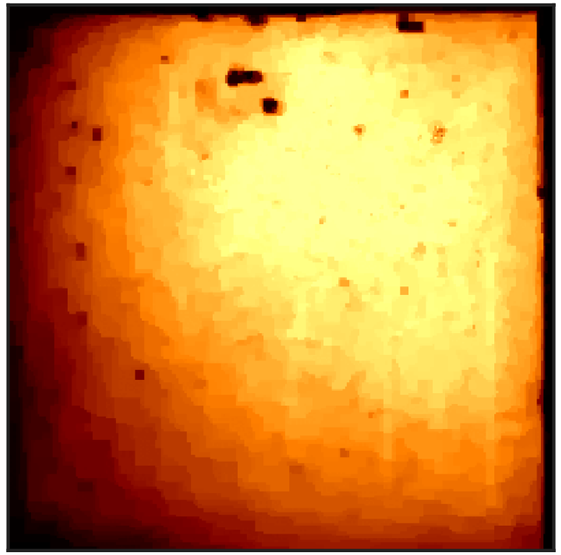
PnP-ADMM-tuned-5
PSNR 32.73 dB | SSIM 0.846
HaarPSI 0.787



C60 (rho=1)
PSNR 29.98 dB | SSIM 0.762
HaarPSI 0.613



C60 optimum (rho=0.43241)
PSNR 30.93 dB | SSIM 0.784
HaarPSI 0.710



Per-image PSNR gain from scaling: +0.947 dB | Compared with PnP-ADMM-tuned-5: -1.802 dB

Ternary rho range [0.0001, 10]; final bracket [0.43072, 0.43748], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

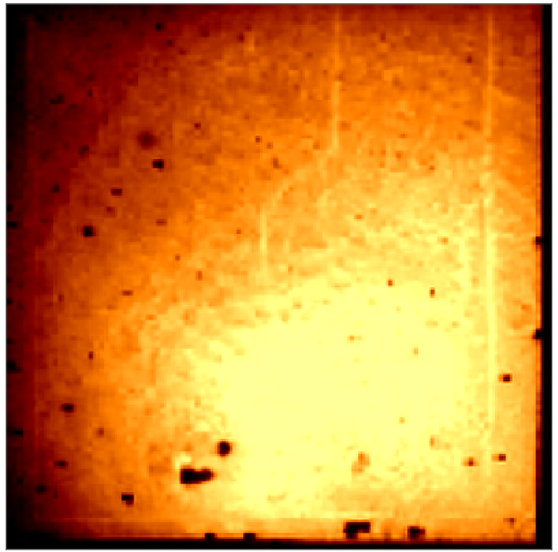


Display [0,1]; metrics on raw images

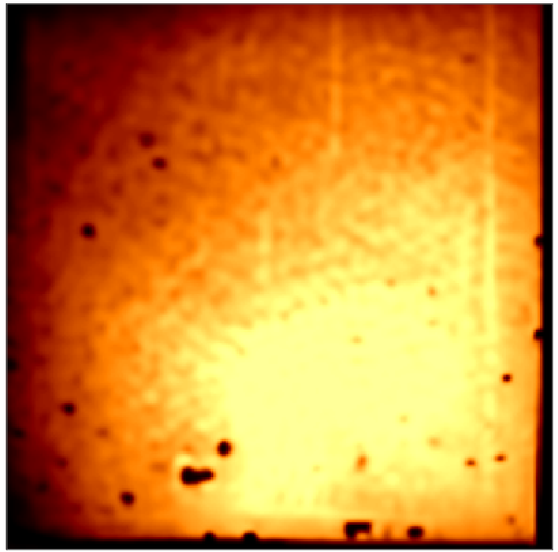
Bad-UTV evaluation | Perovskite_0098

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

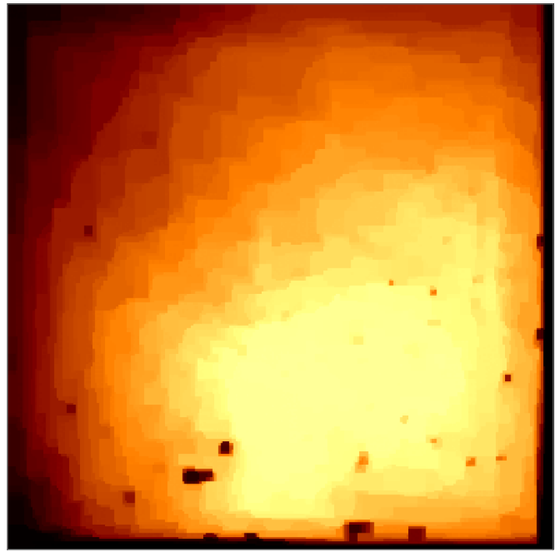
GT
Ground truth



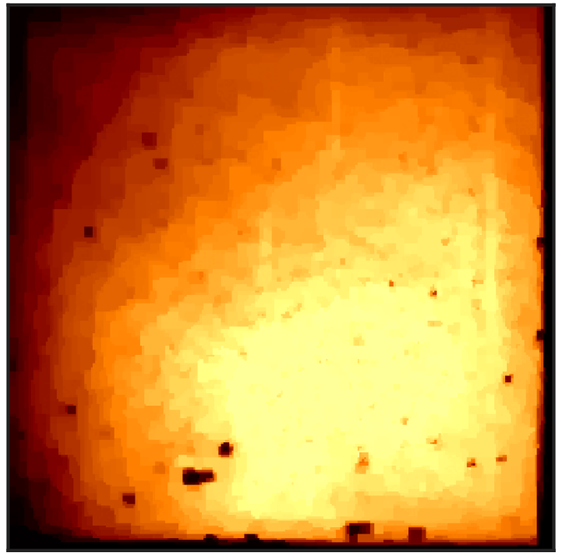
PnP-ADMM-tuned-5
PSNR 32.91 dB | SSIM 0.854
HaarPSI 0.785



C60 (rho=1)
PSNR 30.46 dB | SSIM 0.780
HaarPSI 0.616



C60 optimum (rho=0.44526)
PSNR 31.26 dB | SSIM 0.797
HaarPSI 0.709



Per-image PSNR gain from scaling: +0.800 dB | Compared with PnP-ADMM-tuned-5: -1.655 dB

Ternary rho range [0.0001, 10]; final bracket [0.44356, 0.44695], width 0.00338; 18 iterations, 51 evaluations

UTV C60 loses fine details or contrast

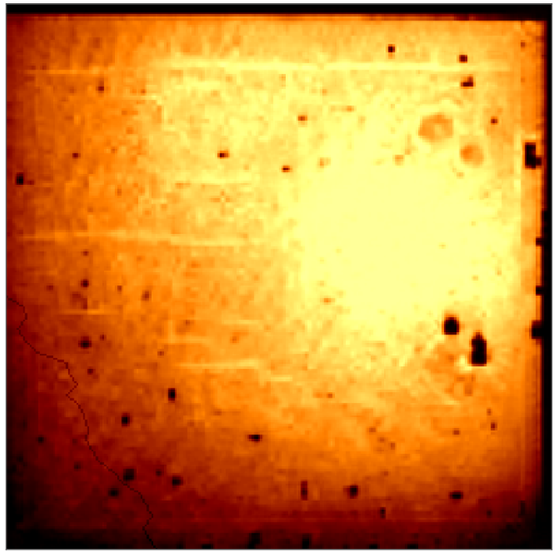


Display [0,1]; metrics on raw images

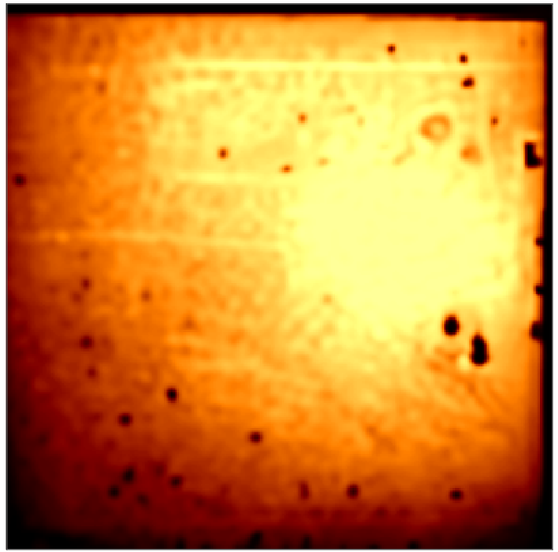
Bad-UTV evaluation | Perovskite_0187

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

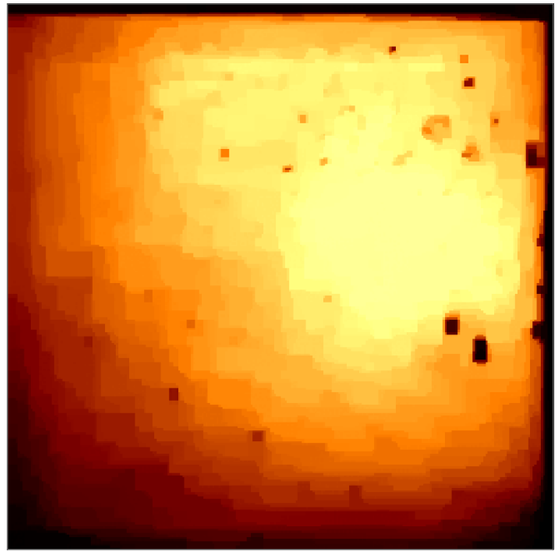
GT
Ground truth



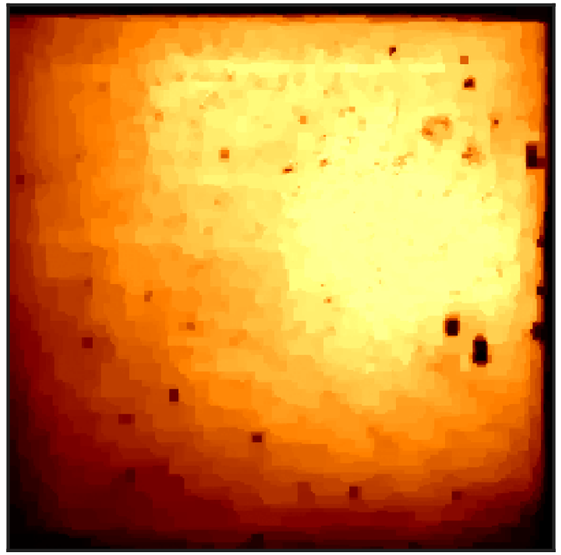
PnP-ADMM-tuned-5
PSNR 32.65 dB | SSIM 0.850
HaarPSI 0.786



C60 (rho=1)
PSNR 29.82 dB | SSIM 0.766
HaarPSI 0.603



C60 optimum (rho=0.45805)
PSNR 30.56 dB | SSIM 0.783
HaarPSI 0.702



Per-image PSNR gain from scaling: +0.739 dB | Compared with PnP-ADMM-tuned-5: -2.084 dB

Ternary rho range [0.0001, 10]; final bracket [0.45466, 0.46143], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

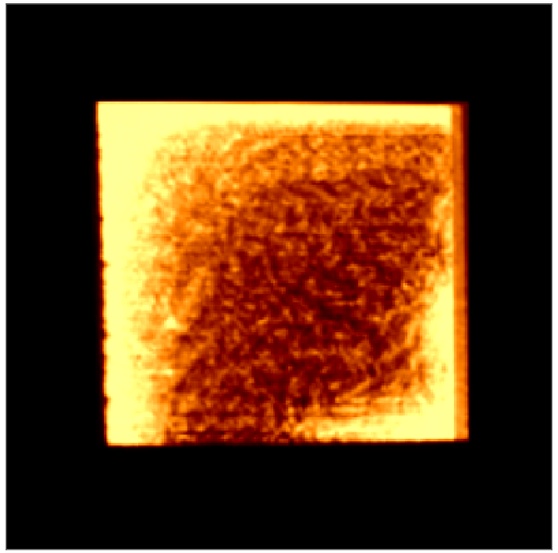


Display [0,1]; metrics on raw images

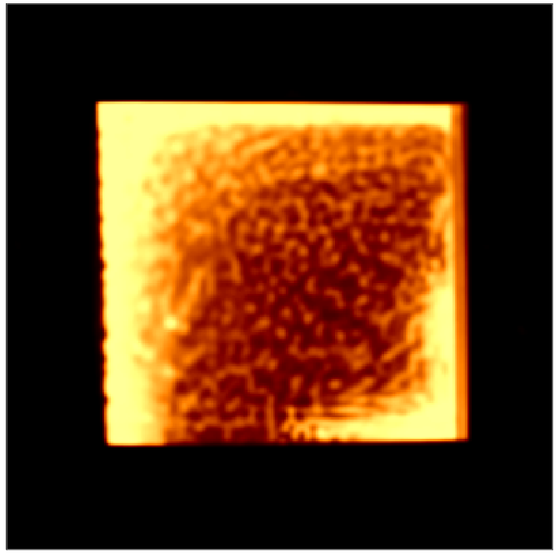
Bad-UTV evaluation | Perovskite_0260

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

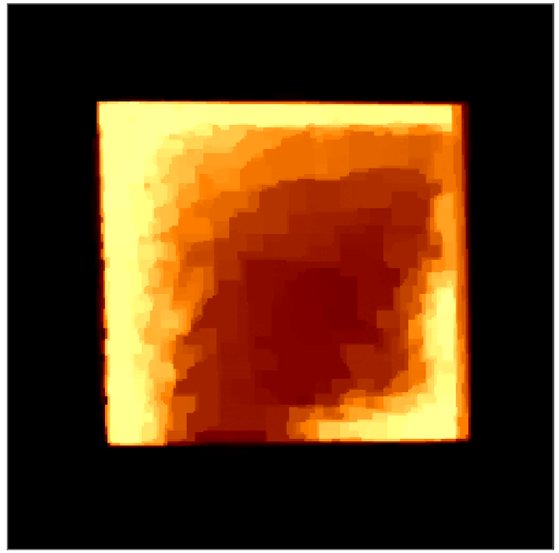
GT
Ground truth



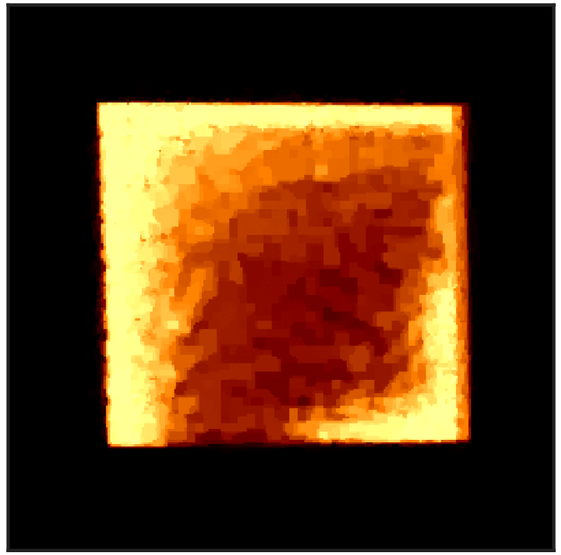
PnP-ADMM-tuned-5
PSNR 32.27 dB | SSIM 0.928
HaarPSI 0.815



C60 (rho=1)
PSNR 28.53 dB | SSIM 0.806
HaarPSI 0.554



C60 optimum (rho=0.33486)
PSNR 29.32 dB | SSIM 0.846
HaarPSI 0.675



Per-image PSNR gain from scaling: +0.795 dB | Compared with PnP-ADMM-tuned-5: -2.953 dB

Ternary rho range [0.0001, 10]; final bracket [0.32979, 0.33656], width 0.00677; 18 iterations, 50 evaluations

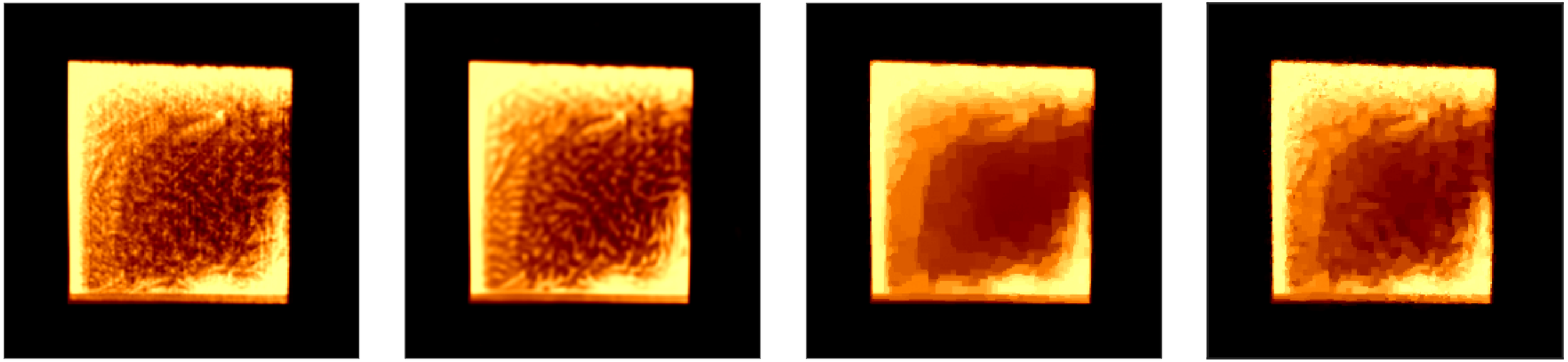
UTV C60 loses fine details or contrast



Display [0,1]; metrics on raw images

Bad-UTV evaluation | Perovskite_0288

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions



Per-image PSNR gain from scaling: +0.612 dB | Compared with PnP-ADMM-tuned-5: -3.136 dB

Ternary rho range [0.0001, 10]; final bracket [0.38189, 0.38865], width 0.00677; 18 iterations, 50 evaluations

UTV C60 loses fine details or contrast

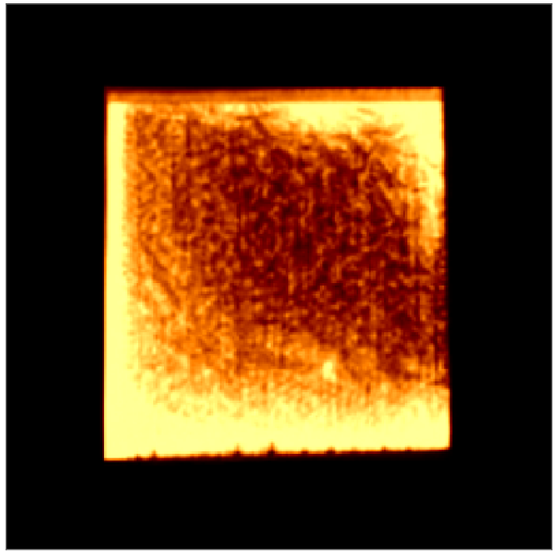


Display [0,1]; metrics on raw images

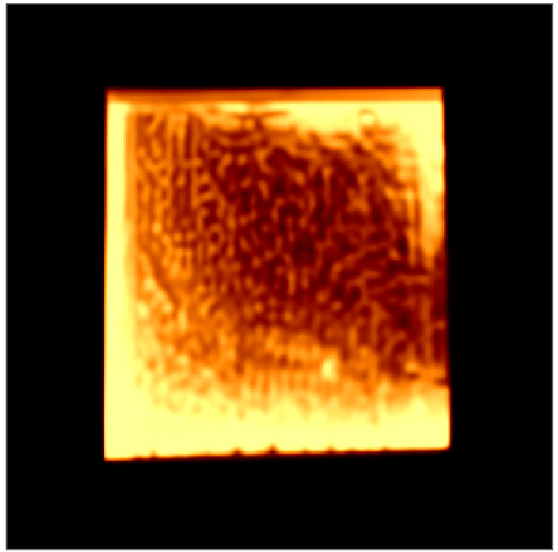
Bad-UTV evaluation | Perovskite_0292

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions

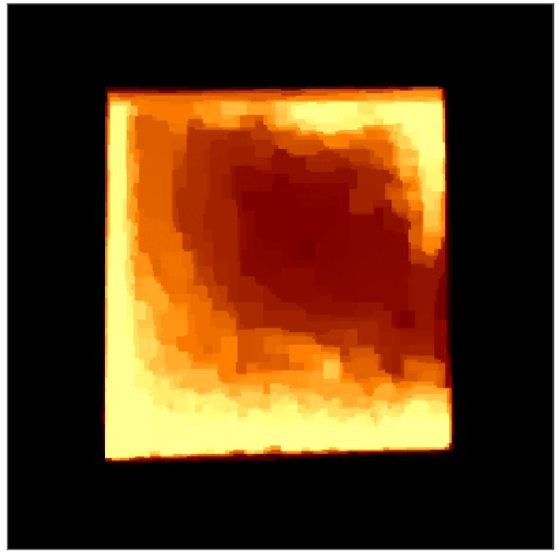
GT
Ground truth



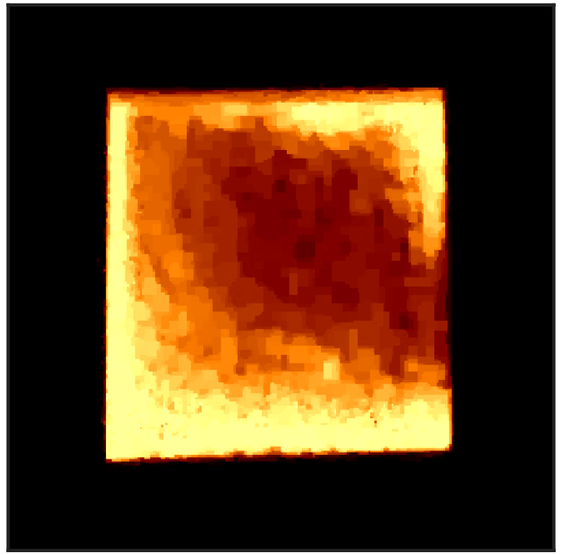
PnP-ADMM-tuned-5
PSNR 32.37 dB | SSIM 0.927
HaarPSI 0.818



C60 (rho=1)
PSNR 28.81 dB | SSIM 0.809
HaarPSI 0.580



C60 optimum (rho=0.38358)
PSNR 29.41 dB | SSIM 0.842
HaarPSI 0.681



Per-image PSNR gain from scaling: +0.597 dB | Compared with PnP-ADMM-tuned-5: -2.961 dB

Ternary rho range [0.0001, 10]; final bracket [0.38019, 0.38696], width 0.00677; 18 iterations, 50 evaluations

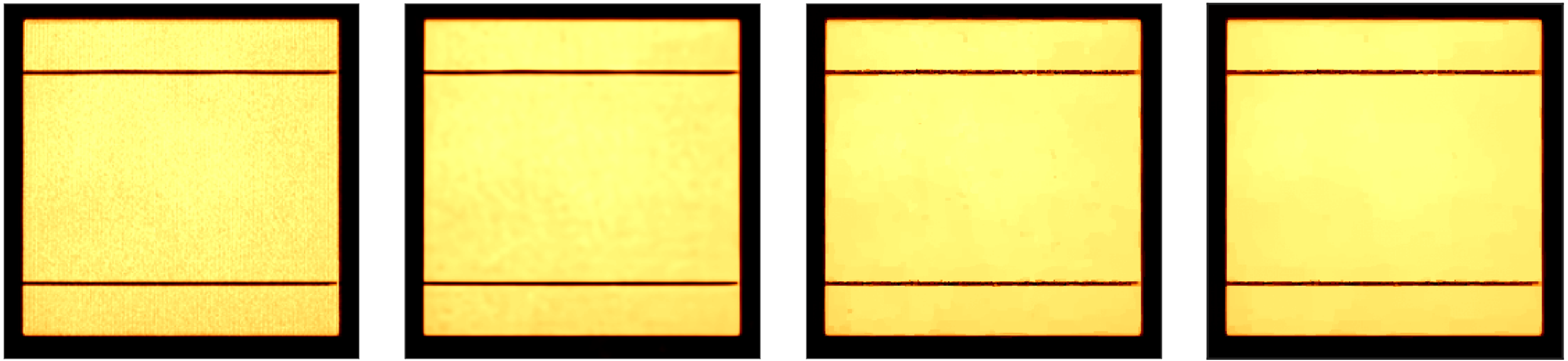
UTV C60 loses fine details or contrast



Display [0,1]; metrics on raw images

Bad-UTV evaluation | Silicone_0767

20% multilevel Hadamard | NPL-default noise | same frozen acquisition for all reconstructions



Per-image PSNR gain from scaling: +0.171 dB | Compared with PnP-ADMM-tuned-5: -1.891 dB

Ternary rho range [0.0001, 10]; final bracket [1.41133, 1.41809], width 0.00677; 18 iterations, 51 evaluations

Visual observation: vertical lines remain in zero-fill but are lost by the other methods.



Display [0,1]; metrics on raw images